



Estimation of pure component properties Part 3. Estimation of the vapor pressure of non-electrolyte organic compounds via group contributions and group interactions

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ABSTRACT

A group contribution method for the estimation of the normal boiling point of non-electrolyte organic compounds, which was published earlier, has been the basis for development of subsequent physical property methods. In this work, the model was extended to enable the prediction of vapor pressure data with special attention to the low-pressure region. The molecular structure of the compound and a reference point, usually the normal boiling point, are the only required inputs and enables the estimation of vapor pressure at other temperatures by group contribution. The structural group definitions are similar to those proposed earlier for the normal boiling point, with minor modifications having been made to improve the predictions. Structural groups were defined in a standardized form and fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions. The new method is based on vapor pressure data for more than 1600 components. The results of the new method are compared to the Antoine correlative equation using parameters stored in the Dortmund Data Bank, as well as, the DIPPR vapor pressure correlations. The group contribution method has proven to be a good predictor, with accuracies comparable to the correlations. Moreover, because the regression of group contributions was performed for a large number of compounds, the results can in several cases be considered more reliable than those of the correlative models that were regressed to individual components only. The range of the method is usually from about the triple or melting point to a reduced temperature of 0.75–0.8.

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1. Introduction

The correlation and prediction of vapor pressures has for long been a very important problem in engineering thermodynamics and has consequently been addressed by many researchers. In early studies, work was usually focused on the pressure region between a few kilopascal and the critical pressure, which is of primary importance for distillation. The description of the temperature dependence of the vapor pressure between a reduced temperature of 0.8 and 1.0 is not trivial.

Correlation or estimation of vapor pressures up to the critical point usually follows either the approach based on the work of Ambrose [1] at the UK. National Physical Laboratory (NPL) based on a Wagner equation form [2] or the Riedel model based on the

Planck–Riedel equation [3,4]. Both models require knowledge of the critical temperature and pressure and reduce the number of coefficients, as well as, their numeric range using physically meaningful constraints. While both models are adequate to correlate experimental vapor pressure data, extrapolation into the low-pressure range is usually unsatisfactory.

Nowadays, a large amount of experimental data for common industrial components is available together with tabulated correlation parameters from different sources, e.g. DIPPR [5], DDB [6], PPDS [7], and there is usually no need to estimate the vapor pressure curve of key components in distillation. Due to the predominant influence of the vapor pressure on the vapor–liquid separation factor, one would also not rely on estimated data for this purpose. In addition, group contribution methods and similar correlations are usually of limited value for small molecules which represent the first members of a homologous series and show the largest deviation from the general trend in the series. Larger molecules in most cases have a low volatility and are less likely purified by distillation. Therefore, the estimation require-

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ments are more for low and medium pressures for fairly complex molecules.

Low and medium vapor pressure estimations for complex large molecules are of great importance:

- in process simulation when it comes to by-products and impurities;
- in environmental protection for the estimation of water–air, soil–air, and the various other compartment distribution coefficients [8];
- in supercritical fluid extraction of solids where these data serve as hypothetical sub-cooled liquid vapor pressures for the estimation of sublimation pressures and
- in various instances where knowledge of the vapor pressure is integral to accurately describe a system or process.

In this work, the focus is on the estimation of vapor pressures below a few atmospheres, in order to avoid the complex dependence on temperature near the critical point. As in our previous work on normal boiling temperatures [9,10] and critical data [11], the resulting larger deviations in the case of small molecules are accepted, as these are difficult anyway to handle by group contribution.

2. General behavior and available methods

The dependence of the logarithm of vapor pressure on reciprocal temperature is given by the well known Clausius–Clapeyron equation

$$\frac{d \ln(P^S/1 \text{ atm})}{d(1/T)} = \frac{-\Delta_{\text{vap}}H}{R\Delta_{\text{vap}}z} \quad (1)$$

When approaching the critical temperature, both $\Delta_{\text{vap}}H$ and $\Delta_{\text{vap}}z$ exhibit a very non-linear and strong change with temperature, which even modern volume translated equations of state find difficult to describe. As discussed in the previous paper [10], the normal boiling point and the critical point are governed by very different physical phenomena. At the normal boiling point the gain in (translational) entropy from vaporization is balanced out by the heat required to overcome the intermolecular interactions in the dense and cool (moderately structured) liquid phase, however, the critical point is determined by the equilibrium between repulsive and attractive forces in a moderately dense unstructured fluid ($dP/dV=0$). Estimation of the critical temperature T_c based on the normal boiling temperature T_b and vice versa is often of limited accuracy as the ratio T_c/T_b depends strongly on the size and chemical nature of the molecule. As an additional complication, between T_b and T_c the slope $d \ln P^S/d(1/T)$ shows a point of inflection due to a minimum in $\Delta_{\text{vap}}H/\Delta_{\text{vap}}z$.

For this reason, corresponding states methods are usually not able to yield a reasonable estimate, unless information on the vapor pressure in the vicinity of the normal boiling temperature is utilized, usually using the acentric factor ω .

Even then they require critical temperature and pressure as a point of reference. Knowledge about the critical point is limited to a few hundred, mostly small molecules. Estimation or correlation techniques requiring the knowledge of the critical point, although very commonly used, will not be considered in this work to avoid unnecessary complications.

Several group contribution methods have been published for the estimation of vapor pressures without requiring knowledge about the critical point [12,13]. Many more can be constructed by combining estimation methods, for example for the normal boiling temperature, virial coefficient and liquid density, the heat of vaporization, and the difference in heat capacity of the two phases.

Several approaches were published to calculate the vapor pressure directly from molecular properties without employing group contribution [14,15].

Another approach uses the UNIFAC method for the prediction of vapor pressures [16,17]. A disadvantage of this method is the requirement for additional data like virial coefficients and the limited range of applicability.

3. Development of the new method and results

The new method employs a slightly improved fragmentation scheme compared to the method for the estimation of normal boiling temperature published earlier [10]. The list of structural groups for the new method, second-order corrections and interacting groups are given in Tables 1–3 respectively, and the group parameter values for vapor pressure estimation are given in Tables 4–6. Structural groups were defined in a standardized form and fragmentation of the molecular structures was performed by an automatic procedure to eliminate any arbitrary assumptions [18].

For a number of groups listed in Table 1, no parameters are available and vapor pressure estimation is not possible. The groups are nevertheless reported. In case data or reasonable estimates for other components containing these groups are available to the user, it is possible to calculate or regress the missing contribution. If the groups would have been discarded from the method, it might still have been possible to perform property estimation using simpler (lower priority) groups, but it would have led to large errors. For interacting groups, not every interaction is of similar importance. Ignoring specific group interactions will lead to similar errors, as demonstrated in the case of the normal boiling point estimation [10].

3.1. Development of the model equation

Assuming an ideal vapor phase and neglecting the small liquid phase volume it follows that

$$\frac{d \ln(P^S/1 \text{ atm})}{d(1/T)} = - \frac{\Delta_{\text{vap}}H(T_0) + \int_{T_0}^T (C_{\text{coex}}^V - C_{\text{coex}}^L) dT}{R} \quad (2)$$

where

$C_{\text{coex}}^V - C_{\text{coex}}^L$ is the difference between the molar heat capacity of the vapor and liquid phase along the vapor–liquid saturation curve.

Assuming that

$$\Delta C_{\text{coex}} = C_{\text{coex}}^V - C_{\text{coex}}^L = \text{constant} \quad (3)$$

leads to the well known vapor pressure equation (Kirchhoff):

$$\ln \left(\frac{P^S}{1 \text{ atm}} \right) = A + \frac{B}{T} + C \ln \left(\frac{1}{T} \right) \quad (4)$$

Unfortunately, this equation cannot be written in temperature-explicit form. To avoid this, it was decided to use the Antoine equation, which is mathematically simpler, but is able to describe a similar curvature at not too high pressures (HPs) and usually extrapolates more reliably to low pressures:

$$\log \left(\frac{P^S}{1 \text{ atm}} \right) = A - \frac{B}{T - C} \quad (5)$$

The disadvantage of the Antoine equation, however, is its discontinuity at $T=C$ and a physically unrealistic increase in the slope of the vapor pressure curve at very low pressures. The value of C was examined by correlating vapor pressure data against the normal boiling point for several hundred components and it was found

Table 1

Group definitions (ID—identification number, PR—priority)

Group	Description	Name	ID PR ^a	Occurs, e.g. in
Periodic group^b 17				
Fluorine				
F–	F– connected to non-aromatic C or Si	F–(C,Si)	19, 87	2-Fluoropropane
	F– connected to C or Si with at least one F or Cl neighbor and one other atom	F–((C,Si)–([F,Cl]))-a	22, 84	Trimethylfluorosilane
	F– connected to C or Si already substituted with at least one F and two other atoms	F–((C,Si)–([F]))-b	21, 81	1-Chloro-1,2,2,2-tetrafluoroethane[R124] Difluoromethylsilane
	F– connected to C or Si already substituted with at least one Cl and two other atoms	F–((C,Si)–(Cl))-b	102, 82	1,1,1-Trifluoroethane 2,2,3,3-Tetrafluoropropionic acid
	F– connected to C or Si already substituted with two F or Cl	F–((C,Si)–([F,Cl] ₂))	23, 83	Trichlorofluoromethane[R11] 1,1-Dichloro-1-fluoroethane [R141B]
	F– connected to an aromatic C	F–(C(a))	24, 86	1,1,1-Trifluorotoluene 2,2,2-Trifluoroethanol, trifluoroacetic acid
	F– on a C=C (vinylfluoride)	F–C=C<	20, 85	Fluorobenzene 4-Fluoroaniline
				Vinyl fluoride Trifluoroethene, perfluoropropylene
Chlorine				
Cl–	Cl– connected to C or Si not already substituted with F or Cl	Cl–(C,Si)	25, 71	Butyl chloride
	Cl– connected to C or Si already substituted with one F or Cl	Cl–((C,Si)–([F,Cl]))	26, 70	2-Chloroethanol, chloroacetic acid Dichloromethane
	Cl– connected to C or Si already substituted with at least two F or Cl	Cl–((C,Si)–([F,Cl] ₂))	27, 68	Dichloroacetic acid, dichlorosilane
	Cl– connected to an aromatic C	Cl–(C(a))	28, 72	Ethyl trichloroacetate Trichloroacetoneitrile
	Cl– on a C=C (vinylchloride)	Cl–C=C<	29, 69	Chlorobenzene Vinyl chloride
COCl–	COCl– connected to C (acid chloride)	COCl–	77, 18	Acetyl chloride Phenylacetic acid chloride
Bromine				
Br–	Br– connected to a non-aromatic C or Si	Br–(C,Si(na))	30, 65	Ethyl bromide Bromoacetone
	Br– connected to an aromatic C	Br–(C(a))	31, 66	Bromobenzene
Iodine				
I–	I– connected to C or Si	I–(C,Si)	32, 63	Ethyl iodide 2-Iodotoluene
Periodic group 16				
Oxygen				
–OH	–OH for aliphatic chains with less than five C (cannot be connected to aromatic groups)	–OH (<C ₅) (z)	36, 92	Ethanol
	–OH connected to C or Si substituted with one C or Si in an at least five C or Si containing chain (primary alkanols)	–OH (>C ₄) (z)	35, 88	Propanediol 1-Nonanol
	–OH connected to a C or Si substituted with two C or Si in at least three C or Si containing chain (secondary alkanols)	HO–((C,Si) ₂ H–(C,Si)–(C,Si))–B4, 90 (z)		Tetrahydrofurfuryl alcohol, ethylene cyanohydrin
	–OH connected to C which has four non-hydrogen neighbors (tertiary alkanols)	–OH <i>tert</i>	33, 91	2-Butanol Cycloheptanol
	–OH connected to an aromatic C (phenols)	HO–(Ca) (z)	37, 89	<i>tert</i> -Butanol Diacetone alcohol
				Phenol
	–O– connected to two neighbors which are each either C or Si (ethers)	(C,Si)–O–(C,Si) (z)	38, 94	Methyl salicylate Diethyl ether
	–O– in an aromatic ring with aromatic C as neighbors	(C(a))–O(a)–(C(a)) (z)	65, 93	1,4-Dioxane Furan
–CHO	CHO– connected to non-aromatic C (aldehydes)	CHO–(Cna) (z)	52, 53	Furfural Acetaldehyde
	CHO– connected to aromatic C (aldehydes)	CHO–(Ca) (z)	90, 52	Pentanedial Furfural
>C=O	–CO– connected to two non-aromatic C (ketones)	O=C<(Cna) ₂ (z)	51, 4	Benzaldehyde Acetone
	–CO– connected to two C with at least one aromatic C (ketones)	(O=C<(C) ₂)a (z)	92, 54	Methyl cyclopropyl ketone Acetophenone
	–CO connected to N	>N–(C=O) (z)	109, 39	Benzophenone Methyl thioacetate
	–CO connected to two N (urea)	>N–(C=O)–N<	100, 3	Urea-1,1,3,3-tetramethyl
	1,2-Diketone (no estimation possible)	O=C–C=O	118, 1	2,3-Butandione
O=C(–O–) ₂	Non-cyclic carbonate diester	O=C(–O–) ₂	79, 14	Dimethyl carbonate
COOH–	–COOH connected to C (carboxylic acid)	COOH–(C) (z)	44, 23	Acetic acid

Table 1 (Continued)

Group	Description	Name	ID PR ^a	Occurs, e.g. in	
	–COO–	HCOO– connected to C (formic acid ester)	HCOO–(C) (z)	46, 26	Ethyl formate Phenyl formate
	–COO–	–COO– connected to two C (ester) in a chain	(C)–COO–(C) (z)	45, 24	Ethyl acetate Vinyl acetate
	–COO–	–COO– in a ring, C is connected to C (lactone)	–C(r)OO– (z)	47, 25	ε-Caprolactone Crotonolactone
	–OCOO–	–CO connected to two O (carbonates)	–OCOO–	103, 33	Propylene carbonate 1,3 Dioxolan-2-one
	–OCON<	–CO connected to O and N (carbamate)	–OCON<	99, 2	Trimethylsilyl methylcarbamate
	>(OC ₂)<	>(OC ₂)<(epoxide)	>(OC ₂)<	39, 50	Propylene oxide
	–CO–O–CO–	Anhydride connected to two C	$\begin{matrix} \text{C}=\text{O}-\text{O}-\text{C}=\text{O}- \\ \text{(z)} \end{matrix}$	76, 11	Acetic anhydride Butyric anhydride
		Cyclic anhydride connected to two C connected by a double bond or aromatic bond	(–C=O–O–C=O–) _r	96, 10	Maleic anhydride Phthalic anhydride
	–O–O–	Peroxide	–O–O–	94, 31	Di- <i>tert</i> -butylperoxide
	Sulphur				
–S–S–	–S–S– (disulfide) connected to two C	(C)–S–S–(C)	55, 51	Dimethyldisulfide 1,2-Dicyclopentyl-1,2-disulfide	
–SH	–SH connected to C (thiols, mercaptanes)	SH–(C) (z)	53, 73	1-Propanethiol	
–S–	–S– connected to two C (thioether)	(C)–S–(C) (z)	54, 74	Methyl ethyl sulfide	
	–S– in an aromatic ring (aromatic thioether)	–S(a)– (z)	56, 75	Thiazole Thiophene	
–SO ₂ –	Non-cyclic sulfone connected to two C (sulfone)	(C)–SO ₂ –(C)	82, 17	Sulfolane divinylsulfone	
>SO ₄	Sulfates	>SO ₄	104, 34	Dimethyl sulfate	
–SO ₂ N<	–S(=O) ₂ connected to N (sulfonamide)	–SO ₂ N<	105, 35	<i>N,N</i> -diethylmethanesulfonamide	
>S=O	Sulfoxide	>S=O	107, 37	1,4-Thioxane- <i>S</i> -oxide Tetramethylene sulfoxide	
SCN–	SCN– (isothiocyanate) connected to C	SCN–(C)	81, 19	Allyl isothiocyanate	
Selenium					
>Se<	>Se< connected to at least one C or Si	>Se<	116, 46	Dimethyl selenide	
Periodic group 15					
Nitrogen					
NH ₂ –	NH ₂ – connected to either C or Si (primary amine)	NH ₂ –(C,Si) (z)	40, 96	Hexylamine Ethylenediamine	
	NH ₂ – connected to an aromatic C (aromatic primary amine)	NH ₂ –(Ca) (z)	41	Aniline Benzidine	
–NH–	–NH– connected to two C or Si neighbors (secondary amine)	(C,Si)–NH–(C,Si) (z)	42, 100	Diethylamine Diallyl amine	
	–NH– connected to two C or Si neighbors in a ring (cyclic secondary amine)	(C,Si) _r –NH–(Ca,Si) _r (z)	97, 99	Morpholine Pyrrolidine	
>N<	>N– connected to three C or Si neighbors (tertiary amine)	(C,Si) ₂ >N–(C,Si)	43, 101	<i>N,N</i> -dimethylaniline Nicotine	
	>N> connected to four C or Si (quartenary amine)	(C,Si) ₂ >N<(C,Si) ₂	101, 32	<i>N,N,N,N</i> -tetramethylmethylenediamine	
=N–	Double bonded amine connected to at least a C or Si	(C,Si)=N–	91, 102	Acetonin	
–N–	Aromatic –N– in a five-membered ring, free electron pair	=N(a)–(r5) (z)	66, 98	Piperidine Thiazole	
=N–	Aromatic =N– in a six-membered ring	=N(a)–(r6) (z)	67, 97	Pyridine Nicotine	
C≡N–	–C≡N (cyano-group) connected to C (cyanide)	(C)–C≡N (z)	57, 55	Acetonitrile 2,2'-Dicyano diethyl sulfide	
	–C≡N (cyano-group) connected to N (cyanamide)	(N)–C≡N	111, 41	Dimethylcyanamide	
	–C≡N (cyano-group) connected to S (thiocyanate)	(S)–C≡N	108, 38	Methyl thiocyanate	
CNCNC-r	Imadizole (aromatic 5-ring)	..=CNC=NC=..	106, 36	1-Methyl-1-imadizole	

Table 1 (Continued)

Group	Description	Name	ID PR ^a	Occurs, e.g. in
-CONH<	-CONH ₂ (amide)	-CONH ₂	50, 27	Acetamide
	-CONH- (monosubstituted amide)	-CONH-	49, 48	N-methylformamide 6-Caprolactam
	-CON< (disubstituted amide)	-CON<	48, 49	N,N-dimethylformamide (DMF)
OCN-	OCN- connected to C or Si (isocyanate)	OCN-(z)	80, 28	Butylisocyanate Hexamethylene diisocyanate
ONC-	ONC- (oxime)	ONC-	75, 29	Methyl ethyl ketoxime
-ON=	-ON= connected to C or Si (isoxazole)	-ON=(C,Si)	115, 45	Isoxazole 5-Phenyl isoxazole
NO ₂ -	Nitrites (esters of nitrous acid)	O=N-O-(C)	74, 22	Ethyl nitrite Nitrous acid methyl ester
	NO ₂ - connected to aliphatic C	NO ₂ -(C)	68, 20	1-Nitropropane
	NO ₂ - connected to aromatic C	NO ₂ -(C(a)) (z)	69, 21	Nitrobenzene
NO ₃ -	Nitrate (esters of nitric acid)	NO ₃ -	72, 13	N-butylnitrate 1,2-Propanediol dinitrate
Phosphorous >P(O-) ₃	Phosphate triester	PO(O-) ₃	73, 9	Triethyl phosphate Tris-(2,4-dimethylphenyl) phosphate
>P<	Phosphorus connected to at least one C or S (phosphine, phosphane)	>P-(C,Si)	113, 43	Triphenylphosphine Triethylphosphane
Arsine AsCl ₂ -	AsCl ₂ connected to C	AsCl ₂ -	84, 16	Ethylarsenic dichloride
Periodic group 14				
Carbon				
-CH ₃	CH ₃ - not connected to either N, O, F or Cl	CH ₃ -(ne)	1, 105	Decane
	CH ₃ - connected to either N, O, F or Cl	CH ₃ -(e)	2, 103	Dimethoxymethane Methyl butyl ether
	CH ₃ - connected to an aromatic atom (not necessarily C)	CH ₃ -(a)	3, 104	Toluene p-Methyl-styrene
-CH ₂ -	-CH ₂ - in a chain	-C(c)H ₂ -	4, 112	Butane
	-CH ₂ - in a ring	-C(r)H ₂ -	9, 113	Cyclopentane
>CH-	>CH- in a chain	>C(c)H-	5, 119	2-Methylpentane
	>CH- in a ring	>C(r)H-	10, 118	Methylcyclohexane
>C<	>C<	>C(c)<	6, 121	Neopentane
	in >C< in a chain connected to at least one aromatic carbon	>C(c)<(a)	8, 109	Ethylbenzene Diphenylmethane
	chain >C< in a chain connected to at least one F, Cl, N or O	>C(c)<(e)	7, 108	Ethanol
	>C< in a ring	>C(r)<	11, 120	Beta-pinene
	>C< in a ring connected to at least one aromatic carbon	>C(r)<(Ca)	14, 107	Indene 2-Methyl tetralin
	>C< in a ring connected to, at least one N or O which are not part of the ring, or one Cl or F	>C(r)<(e,c)	12, 110	Cyclopentanol Menthol
	>C< in a ring connected to at least one N or O which are part of the ring	>C(r)<(e,r)	13, 111	Morpholine Nicotine
	Aromatic =CH-	=C(a)H-	15, 106	Benzene
=C(a)<	Aromatic =C< not connected to either O, N, Cl or F	=C(a)<(ne)	16, 117	Ethylbenzene Benzaldehyde
	Aromatic =C< with three aromatic neighbors and three aromatic bonds	(a)=C(a)< ₂ (a)	18, 115	Naphthalene Quinoline
	Aromatic =C< connected to either O, N, Cl or F	=C(a)<(e)	17, 114	Aniline Phenol
	Aromatic =C< with three aromatic neighbors and two aromatic bonds (aliphatic bridge bond between aromatic rings)	C(a)=C(a)< ₂ (a) (bridge)	214, 116	Biphenyl m-Terphenyl
>C=C<	H ₂ C=C<(1-ene)	H ₂ C(c)=C<	61, 57	1-Hexene

Table 1 (Continued)

Group	Description	Name	ID PR ^a	Occurs, e.g. in
	>C=C< (both C have at least one non-H neighbor)	>C(c)=C(c)<	58, 62	2-Heptene Mesityl oxide
	Non-cyclic >C=C< connected to at least one aromatic C	>C(c)=C(c)<(C(a))	59, 59	Isosafrole Cinnamic alcohol
	Cyclic >C=C<	>C(r)=C(r)<	62, 60	Cyclopentadiene
	Non-cyclic >C=C< substituted with at least one F, Cl, N or O	-(e)C(c)=C(c)<	60, 58	<i>trans</i> -1,2-Dichloroethylene Perfluoroisoprene
$\text{-C}\equiv\text{C-}$	$\text{HC}\equiv\text{C-}$ (1-ine)	$\text{HC}\equiv\text{C-}$	64, 56	1-Heptyne
	$\text{-C}\equiv\text{C-}$ with two non-H neighbors	$\text{-C}\equiv\text{C-}$	63, 61	2-Octyne
>C=C=C<	Two cumulated double bonds	>C=C=C<	87, 5	1,2-Butadiene Dimethyl allene
>C=C-C=C<	Two conjugated double bonds in a ring	>C=C-C=C<	88, 6	Cyclopentadiene Abietic acid
>C=C-C=C<	Two conjugated double bonds in a chain	>C=C-C=C<	89, 7	Isoprene 1,3-Hexadiene
$\text{-C}\equiv\text{C-C}\equiv\text{C-}$	Two conjugated triple bonds	$\text{-C}\equiv\text{C-C}\equiv\text{C-}$	95, 8	2,4-Hexadiyne
Silicon				
>Si<	>Si<	>Si<	70, 80	Butylsilane
	>Si< attached to no carbon or hydrogen	>Si<(C,H)_0	216, 76	Tetrachlorosilane Tetramethoxysilane
	>Si< attached to one carbon or hydrogen	>Si<(C,H)_1	215, 77	Trichlorosilane Methyltrichlorosilane
	>Si< attached to two carbon or hydrogen	>Si<(C,H)_2	93, 78	Dichlorodimethylsilane Dichlorodiethylsilane
	>Si< attached to three carbon or hydrogen	>Si<(C,H)_3	71, 79	Hexamethyl disiloxane
Germanium				
>Ge<	>Ge< connected to four carbons	$\text{(C)}_2\text{>Ge<(C)}_2$	86, 67	Tetramethylgermane
$\text{GeCl}_3\text{-}$	$\text{GeCl}_3\text{-}$ connected to carbons	$\text{GeCl}_3\text{-}$	85, 12	Fluorodimethylsilyl (trichlorogermanyl)methane
Tin				
>Sn<	>Sn< connected to four carbons	$\text{(C)}_2\text{>Sn<(C)}_2$	83, 64	Tetramethylstannane
Periodic group 13				
Boron				
B(O-)_3	Non-cyclic boric acid ester	B(O-)_3	78, 15	Triethyl borate
Aluminum				
>Al<	>Al< connected to at least one C or Si	>Al<	117, 47	Triethylaluminum

Abbreviations: (e), very electronegative neighbours (N, O, F, Cl); (ne), not very electronegative neighbours (not N, O, F, Cl); (na), non-aromatic atom or neighbour; (a), aromatic atom or neighbour; (c), atom or neighbour is part of a chain; (r), atom or neighbour is part of a ring; (z), groups considered to be non-additive.

^a Priority numbers define the importance of a group. If two different groups can be used, the one with the lower priority number must be used.

^b Periodic group number of the most significant element in the structural group. Periodic group numbers are used as defined in the IUPAC version of the periodic table of elements.

that there was a correlation with the normal boiling point temperature, with $C \approx Tb/8$. In 1946, Thomson [19] proposed the expression, $C = -18 + 0.19 \times T_b$, based on a much smaller set of data. By relating C to the normal boiling point temperature the number of parameters is reduced to two and the discontinuity is always in the range of $T_r \approx 0.08$ and of no concern for practical application. Introducing the normal boiling point as one of the parameters, by replacing T by $T_{rb} = T/T_b$ and defining $B = 4.1012 + dB$ leads to

$$\log \left(\frac{P^S}{1 \text{ atm}} \right) = (4.1012 + dB) \left(\frac{T_{rb} - 1}{T_{rb} - (1/8)} \right) \quad (6)$$

In this equation, the value of dB does not depend significantly on the normal boiling point, nor on the units of pressure used. The constant, 4.1012, was computed as a mean value from the correlation of vapor pressure data for several hundred nonpolar components. In this form, the equation can also be applied as an approximation when using $dB = 0$. The value of dB can be estimated using the approach presented in this paper or via an "educated guess" as it is directly correlated with the strength of the intermolecular forces in the mixture. dB is typically in the range from -0.5 to 2 and is close to zero for nonpolar components.

The advantage of this equation is that the parameters relate directly to the absolute value (normal boiling point) and slope of

Table 2
Second-order groups and corrections

Name contribution (K)	Description	ID	Example (s)
C=C—C=O	—C=O connected to sp ² carbon	134	Benzaldehyde Furfural
(C=O)—C([F,Cl] _{2,3})	Carbonyl connected to carbon with two or more halogens	119	Dichloroacetyl chloride
(C=O)—C([F,Cl] _{2,3}) ₂	Carbonyl connected to two carbon with two or more halogens each	120	Perfluoro-2-propanone
C—[F,Cl] ₃	Carbon with three halogens	121	1,1,1-Trifluorotoluene
(C) ₂ —C—[F,Cl] ₂	Secondary carbon with two halogens	122	2,2-Dichloropropane
no hydrogen	Component has no hydrogen	123	Perfluoro compounds
one hydrogen	Component has one hydrogen	124	Nonafluorobutane
3/4 ring	A three or four-membered non-aromatic ring	125	Cyclobutene
5 ring	A five-membered non-aromatic ring	126	Cyclopentane
ortho pair(s)	Ortho position—counted only once and only if there are no meta or para pairs	127	o-Xylene
meta pair(s)	Meta position—counted only once and only if there are no para or ortho pairs	128	m-Xylene
para pair(s)	Para position—counted only once and only if there are no meta or ortho pairs	129	p-Xylene
((C=)(C)C—CC ₃)	Carbon–carbon bond with four single bonded and one double bonded carbon neighbor	130	tert-Butylbenzene
C ₂ C—CC ₂	Carbon–carbon bond with four carbon neighbors, two on each side	131	Bicyclohexyl
C ₃ C—CC ₂	Carbon–carbon bond with five carbon neighbors	132	Ethyl bornyl ether
C ₃ C—CC ₃	Carbon–carbon bond with six carbon neighbors	133	2,2,3,3-Tetramethylbutane
Si< (F, Cl, Br, I)	A silicon attached to a halogen atom	217	Trichloroethylsilane

the vapor pressure curve. Both values can easily be derived from experimental data as a basis for group contribution method development.

Using Eq. (6), expressions for further properties (boiling temperature and enthalpy of vaporization) can be derived and are given in Appendix A.

In order to calculate the value of the parameter dB, the following equation is used:

$$dB = \left(\sum_i N_i C_i + GI \right) - 0.176055 \quad (7)$$

where N_i is the number of groups of type i , C_i the group contribution of group i , GI the total group interaction contribution (see Eq. (8)).

As in the previous method, a group interaction contribution for non-additive groups is adopted here. The total group interaction contribution can be calculated from the following equation:

$$GI = \frac{1}{n} \sum_{i=1}^m \sum_{j=1}^m \frac{C_{i-j}}{m-1} \quad \text{where} \quad C_{i-j} = C_{j-i} \quad (8)$$

with C_{i-j} , group interaction contribution between group i and group j [K] ($C_{i-i} = 0$), n number of atoms (except hydrogen), m total number of interaction groups in the molecule

3.2. Hydrocarbon compounds

As in the previous studies [10,11], model development began with regression of group contributions for saturated, unsaturated, and aromatic hydrocarbons and their subclasses of components. This strategy is employed to detect unreliable data and flaws in the group contribution approach. The final results presented in this

work are based on regression of all components and not just of the individual classes of components.

The results and quality analysis for hydrocarbons are presented in Table 7. For the comparison of the proposed work to the correlative model, it is essential that the same applicable pressure range is employed. Vapor pressure data close to the normal boiling point (within ± 10 –15 K) were not used, due to the divergence of dB in this range. This divergence is caused by the fact that individual dB-values for each data point were calculated from the data point and a reliable normal boiling point. These slope values are strongly affected by experimental errors in the case of small temperature differences.

During the regression of Antoine parameters stored in the DDB, questionable data had been removed and it was decided to compare the estimation results to the results of the correlation equation. The deviations in the case of the correlation equation represent the effect of scatter in the experimental data.

A series plot for the vapor pressure estimation in the case of n -alkanes is presented in Fig. 1. For all compounds, the estimations from this work are in excellent agreement with the experimental data.

3.3. Mono-functional compounds

The definition of mono-functional compounds employed in this work is a set of compounds with a hydrocarbon backbone and only one type of functional group, for example —OH, —NH₂, etc., which has a frequency of one. The group contribution approach is based on the additivity of group increments with respect to the estimated property. In the case of hydrogen-bonding or otherwise associating groups the assumption of simple additivity is not entirely valid. This was also observed during the development of the normal boiling

Table 3
Groups considered to be non-additive (group-ID(s) given in brackets)

Group abbreviation	Group name (group-ID (s))	Group abbreviation	Group name (group-ID (s))
OH	Alkanol (—OH) (34,35,36)	Ats	Aromatic sulphur (—S(a)—) (56)
OH(a)	Phenol (—OH(a)) (37)	SH	Thiol (—SH) (53)
COOH	Carboxylic acid (—COOH) (44)	NH2	Primary amine (—NH ₂) (40, 41)
EtherO	Ether (—O—) (38)	NH	Secondary amine (>NH) (42,97)
Epoxy	Epoxide ($\begin{smallmatrix} \diagup \text{OC}_2 \diagdown \end{smallmatrix}$) (39)	OCN	Isocyanate (—OCN) (80)
Ester	Ester (—COOC—) (45,46,47)	CN	Cyanide (—CN) (57)
Ketone	Ketone (—CO—) (51,92)	Nitro	Nitrate (69)
Alde	Aldehyde (—CHO) (52,90)	AN5	Aromatic N in 5-ring (=N(a)—(r5)) (66)
AO	Aromatic oxygen (—O(a)—) (65)	AN6	Aromatic N in 6-ring (=N(a)—(r6)) (67)
Teth	Sulfide (thioether) (—S(na)—) (54)		

Table 4

Vapor pressure curve slope (dB) group contributions, number of components used for regressing these values and deviations for these components

Group ID	Group contribution, dB _i ($\times 10^3$)	Number of components	Mean absolute error	Mean relative error (%)	Standard deviation
1	13.3063	1031	0.0888	1.95	0.1255
2	91.8000	121	0.1171	2.50	0.1571
3	50.1939	109	0.0900	1.99	0.1159
4	54.6564	653	0.0916	1.96	0.1304
5	45.7437	143	0.0820	1.81	0.1144
6	−31.7531	52	0.0743	1.68	0.1092
7	37.8485	549	0.1186	2.54	0.1619
8	96.1386	84	0.1068	2.30	0.1392
9	22.2573	207	0.0812	1.85	0.1232
10	32.8162	98	0.0777	1.81	0.1190
11	4.8500	22	0.0778	1.83	0.0991
12	23.6411	44	0.1196	2.64	0.1555
13	49.8237	51	0.0960	2.16	0.1349
14	−3.6950	15	0.0618	1.39	0.0767
15	32.7177	368	0.0921	2.03	0.1274
16	69.8796	276	0.0936	2.05	0.1321
17	41.5534	142	0.0878	1.94	0.1161
18	43.7191	46	0.1007	2.19	0.1372
19	79.5429	33	0.0827	1.89	0.1052
20	51.2880	6	0.1386	3.18	0.1896
21	42.0887	120	0.1050	2.42	0.1475
22	56.9998	20	0.0964	2.20	0.1539
23	142.1060	3	0.0597	1.42	0.0780
24	45.9652	21	0.0605	1.40	0.0797
25	93.6679	61	0.0896	2.01	0.1223
26	67.8082	30	0.0646	1.52	0.0798
27	55.9304	43	0.0850	2.01	0.1291
28	46.0435	47	0.0982	2.22	0.1299
29	84.9162	20	0.0989	2.29	0.1356
30	104.9291	33	0.0960	2.28	0.1294
31	−40.1837	10	0.1264	3.13	0.1682
32	134.3501	12	0.0669	1.56	0.0836
33	719.3666	27	0.1065	2.18	0.1426
34	758.4218	42	0.1667	3.32	0.1995
35	700.7226	57	0.1716	3.37	0.2059
36	756.0824	29	0.1733	3.38	0.2238
37	441.8437	30	0.0990	2.09	0.1257
38	108.4964	169	0.0982	2.17	0.1401
39	286.9731	12	0.0840	1.91	0.1292
40	251.9212	36	0.1340	2.78	0.1811
41	361.7760	18	0.0810	1.74	0.0939
42	193.7667	20	0.1322	2.85	0.1821
43	−102.7252	29	0.1283	2.85	0.1642
44	1074.1000	35	0.1544	3.00	0.2020
45	355.7381	130	0.1192	2.46	0.1689
46	350.5184	12	0.0881	1.92	0.1209
47	292.8046	2	0.0454	1.05	0.0454
48	269.2471	9	0.1460	3.34	0.1665
49	736.9540	4	0.1193	2.51	0.1288
50	1216.0700	1			
51	255.8480	55	0.1017	2.23	0.1522
52	252.9059	22	0.0824	1.89	0.1119
53	123.2143	41	0.0803	1.81	0.1075
54	127.3380	30	0.0536	1.24	0.0810
55	222.2789	4	0.0564	1.31	0.0585
56	20.1604	13	0.1154	2.68	0.1403
57	226.1873	33	0.0708	1.56	0.1091
58	86.4601	87	0.0566	1.26	0.0870
59	224.2238	4	0.0295	0.61	0.0362
60	134.9382	18	0.1155	2.62	0.1626
61	34.2541	121	0.0776	1.81	0.1024
62	97.4210	55	0.0863	2.00	0.1345
63	206.6665	22	0.0599	1.34	0.0777
64	128.0247	16	0.0832	1.90	0.0960
65	48.8839	8	0.0341	0.80	0.0553
66	375.0486	11	0.1768	3.80	0.1976
67	126.3340	30	0.0785	1.78	0.1082
68	375.8217	3	0.0148	0.34	0.0157
69	238.2066	13	0.1105	2.43	0.1231
70	2.8992	25	0.0969	2.38	0.1308
71	9.3624	28	0.1135	2.56	0.1450
72	603.5347	4	0.2002	4.24	0.2051
73	662.0582	3	0.1739	3.48	0.1852

Table 4 (Continued)

Group ID	Group contribution, $\text{dB}_i (\times 10^3)$	Number of components	Mean absolute error	Mean relative error (%)	Standard deviation
74	510.9666	2	0.0940	2.07	0.0940
75	1317.4360	6	0.0345	0.65	0.0479
76	681.3525	4	0.1201	2.52	0.1287
77	564.1116	7	0.3308	6.90	0.3833
78	391.3697	6	0.0828	1.81	0.1091
79	318.2350	2	0.0060	0.14	0.0060
80	435.8446	16	0.2035	4.45	0.2279
81	218.6012	3	0.0338	0.80	0.0361
82	381.5945	2	0.0093	0.21	0.0093
83	80.2735	3	0.0043	0.11	0.0048
84	231.3919	6	0.0386	0.91	0.0453
85	186.9204	3	0.0029	0.07	0.0032
86	48.5026	1			
87	168.3007	4	0.0295	0.71	0.0361
88	108.5277	5	0.0803	1.95	0.1001
89	213.7165	11	0.0976	2.31	0.1165
90	183.1130	4	0.0341	0.79	0.0482
91	1178.1950 ^a	1			
92	158.3258	9	0.1226	2.75	0.1432
93	−47.3420	27	0.0806	1.81	0.0976
94	186.7950	1			
95	392.2006	1			
96	595.1778	2	0.0389	0.86	0.0389
97	268.7081	11	0.0871	1.96	0.1211
98	183.3467	3	0.0654	1.48	0.0704
99	612.9546	2	0.2161	4.57	0.2161
100	258.9924	1			
101	−316.4392	3	0.2033	5.63	0.2180
102	64.0566	8	0.0467	1.09	0.0740
103	660.2247	2	0.2061	4.41	0.2061
104	554.7941	2	0.0732	1.58	0.0732
105	420.7591	3	0.0950	2.12	0.1096
107	−237.2124	1			
110	37.0590	4	0.1167	2.72	0.1431
111	319.4879	1			
113	118.8412	3	0.0737	1.72	0.0842
115	305.1341	1			
116	191.5058	1			
117	423.5251	1			
118 (do not estimate)					
214	86.9450	11	0.0907	2.00	0.1121
215	−66.5670	11	0.0823	1.93	0.1141
216	−81.1543	8	0.1083	2.44	0.1625

^a Questionable group contribution values.

point and critical property methods. Thus, model development proceeded by regressing data for a functional group, a class or subclass of compounds. The approach was also to analyse the performance of each group, test their predictive capability, and the assumption of simple additivity. As with the hydrocarbon compounds, the final

results presented here are based on a simultaneous regression of data for all components.

For mono-functional alcohol compounds, a systematic deviation was observed at low pressures. For this reason, Eq. (9), which is only applicable to mono-functional alcohol compounds (i.e. a hydrocar-

Table 5

Vapor pressure curve slope (dB) second-order contributions, number of components used for regressing these values and deviations for these components

Group ID	Group contribution, $\text{dB}_i (\times 10^3)$	Number of components	Mean absolute error	Mean relative error (%)	Standard deviation
119	34.3545	14	0.1505	3.23	0.2050
120	2.5030	2	0.0373	0.88	0.0373
121	−83.3326	115	0.0949	2.18	0.1344
122	−64.4854	53	0.1180	2.67	0.1533
123	−125.9208	85	0.1014	2.38	0.1377
124	−47.2962	29	0.0776	1.78	0.1262
125	33.9765	37	0.0808	1.91	0.1107
126	−7.0982	106	0.0756	1.73	0.1106
127	−45.0531	89	0.0883	1.98	0.1203
128	−3.2036	56	0.1034	2.22	0.1571
130	−20.6706	9	0.0598	1.28	0.0819
131	−36.3170	47	0.0607	1.44	0.0949
132	−1.1994	20	0.0785	1.82	0.1011
133	123.7433	4	0.1141	2.57	0.1245
134	−15.9694	59	0.1123	2.37	0.1795
217	36.7574	31	0.0845	2.00	0.1177

Table 6
Vapor pressure curve slope (dB) group interaction contributions, interacting groups, number of components used for regressing these values and deviations for these components

Group ID	Interacting groups	Group contribution, dB _i ($\times 10^3$)	Number of components	Mean absolute error	Mean relative error (%)	Standard deviation
135	OH–OH	–561.5153	20	0.2038	3.69	0.2533
136	OH–NH ₂	1067.6660	3	0.1928	3.46	0.2376
137	OH–NH	42.4825	3	0.2236	4.52	0.2642
140	OH–etherO	–799.5332	23	0.1389	2.81	0.1747
141	OH–epox	–618.2760	1			
142	OH–ester	–1797.6930	2	0.1662	3.58	0.1666
143	OH–ketone	–1181.5990	4	0.1160	2.46	0.1419
145	OH–CN	1431.2430	1			
146	OH–AO	1132.0400	1			
148	OH(a)–OH(a)	–97.6205	2	0.1184	2.32	0.1207
151	OH(a)–etherO	–751.6676	3	0.0169	0.35	0.0191
152	OH(a)–ester	548.4352	1			
157	NH ₂ –NH ₂	1085.8320	7	0.2048	4.03	0.2821
158	NH ₂ –NH	–206.7811	2	0.1840	3.87	0.1854
159	NH ₂ –etherO	–198.2791	1			
160	NH ₂ –ester	–1676.4770	1			
162	NH ₂ –nitro	1659.0340	2	0.1208	2.40	0.1208
165	NH–NH	–307.1018	2	0.0094	0.21	0.0095
166	NH–etherO	65.4421	1			
170	SH–SH	240.3037	10	0.0919	1.93	0.1153
172	COOH–COOH	–2601.7090	2	0.0326	0.55	0.0327
175	COOH–ketone	–787.8563	1			
176	OCN–OCN	–3929.1300	3	0.2014	4.30	0.2196
178	EtherO–etherO	144.6074	61	0.1253	2.71	0.1745
179	EtherO–epox	1118.9580	1			
180	EtherO–ester	–225.7802	7	0.0779	1.73	0.1007
181	EtherO–ketone	1981.2980	1			
182	EtherO–alde	362.7540	1			
183	EtherO–teth	–1425.0170	1			
185	EtherO–CN	743.3353	2	0.0494	1.06	0.0495
187	Epox–epox	–3748.8180 ^a	1			
189	Ester–ester	920.3138	24	0.2180	4.27	0.2660
190	Ester–ketone	1594.1640	2	0.3272	6.56	0.3278
192	Ester–CN	108.1305	2	0.3587	7.83	0.3595
193	Ester–AO	1590.3210	1			
194	Ketone–ketone	–1270.0830	3	0.1858	4.27	0.1934
201	Alde–alde	946.7309	1			
204	Alde–AO	705.3049	1			
205	Teth–teth	838.3372	1			
206	Nitro–nitro	–1501.3550	2	0.1863	3.77	0.1973
208	AtS–AN5	675.0414	2	0.2317	5.07	0.2410
209	CN–AN6	994.4996	1			
210	AO–AN5	135.5896	1			
212	AN6–AN6	–29.6785	3	0.1560	3.54	0.1767
218	COOH–NH ₂	Do not estimate	–			

^a Questionable group contribution values.

Table 7
Vapor pressure average absolute deviations of this work compared to the Antoine equation for the different types of hydrocarbons (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
Hydrocarbons (HC)	424	29.1 ³⁵⁸	6.7 ⁵⁵⁵⁰	1.9 ¹¹⁶⁶²	4.1 ³⁶³⁷	4.0 ²¹²⁰⁷	415	40.3 ³⁵⁵	5.9 ⁵⁵¹²	1.0 ¹¹⁶⁴⁴	3.7 ³⁶³⁶	3.4 ²¹¹⁴⁷
Saturated HC	131	23.1 ¹¹³	6.6 ²¹⁰⁷	1.9 ⁵³⁴⁷	4.7 ²²⁸⁹	3.8 ⁹⁸⁵⁶	131	53.3 ¹¹³	5.5 ²¹⁰⁷	0.6 ⁵³⁴⁷	2.7 ²²⁸⁹	2.7 ⁹⁸⁵⁶
Non-aromatic HC	319	28.7 ¹⁵⁰	6.9 ²⁸⁹¹	2.0 ⁷⁶⁶⁹	4.5 ²⁶⁹⁷	3.8 ¹³⁴⁰⁷	315	47.8 ¹⁵⁰	5.6 ²⁸⁷⁰	0.8 ⁷⁶⁶³	2.7 ²⁶⁹⁷	2.7 ¹³³⁸⁰
Unsaturated HC	188	45.8 ³⁷	7.5 ⁷⁸⁴	2.2 ²³²²	3.3 ⁴⁰⁸	4.0 ³⁵⁵¹	184	31.0 ³⁷	5.9 ⁷⁶³	1.4 ²³¹⁶	2.4 ⁴⁰⁸	2.8 ³⁵²⁴
n-Alkanes	26	24.5 ⁸⁷	7.9 ¹¹³⁶	2.3 ²²¹⁶	5.6 ¹²⁶⁹	5.0 ⁴⁷⁰⁸	26	64.8 ⁸⁷	7.0 ¹¹³⁶	0.5 ²²¹⁶	2.6 ¹²⁶⁹	3.8 ⁴⁷⁰⁸
Alkanes (non-cyclic)	73	24.3 ⁹⁴	7.4 ¹⁵⁴⁵	2.2 ³⁵²⁴	5.0 ²⁰³⁰	4.4 ⁷¹⁹³	73	61.7 ⁹⁴	6.7 ¹⁵⁴⁵	0.6 ³⁵²⁴	2.7 ²⁰³⁰	3.3 ⁷¹⁹³
Alkanes (cyclic)	58	17.6 ¹⁹	4.6 ⁵⁶²	1.4 ¹⁸²³	2.4 ²⁵⁹	2.3 ²⁶⁶³	58	11.8 ¹⁹	2.0 ⁵⁶²	0.5 ¹⁸²³	3.1 ²⁵⁹	1.1 ²⁶⁶³
Aromatic HC	105	29.4 ²⁰⁸	6.6 ²⁶⁵⁹	1.8 ³⁹⁹³	3.0 ⁹⁴⁰	4.3 ⁷⁸⁰⁰	100	34.8 ²⁰⁵	6.2 ²⁶⁴²	1.2 ³⁹⁸¹	6.5 ⁹³⁹	4.5 ⁷⁷⁶⁷
Fused aromatics	31	33.9 ⁶⁸	6.1 ⁶⁶⁶	1.9 ⁸⁶⁸	3.6 ⁷⁵	4.9 ¹⁶⁷⁷	31	31.3 ⁶⁸	10.4 ⁶⁶⁶	2.4 ⁸⁶⁸	18.9 ⁷⁵	7.5 ¹⁶⁷⁷
Alkenes (HC)	139	45.8 ³⁷	7.9 ⁵⁶⁵	2.1 ¹⁸⁰⁶	2.8 ³⁷²	4.0 ²⁷⁸⁰	136	31.0 ³⁷	6.8 ⁵⁵⁶	1.4 ¹⁸⁰³	2.3 ³⁷²	3.0 ²⁷⁶⁸
Alkenes (cyclic) HC	39	22.4 ²	8.9 ²⁶⁵	2.4 ⁵⁷³	2.4 ²¹	4.5 ⁸⁶¹	38	47.5 ²	9.0 ²⁶³	2.0 ⁵⁷²	4.3 ²¹	4.3 ⁸⁵⁸
Alkynes (HC)	32	–	6.2 ¹⁰⁸	2.5 ³³³	9.5 ³³	3.8 ⁴⁷⁴	32	–	5.7 ¹⁰⁸	1.4 ³³³	3.9 ³³	2.6 ⁴⁷⁴

NC, number of components; AV, average relative deviations for all points. The following abbreviations denote pressure ranges, ELP— $P < 0.01$ kPa, LP— 0.01 kPa $\leq P \leq 10$ kPa, MP— 10 kPa $< P \leq 500$ kPa, HP— $P > 500$ kPa

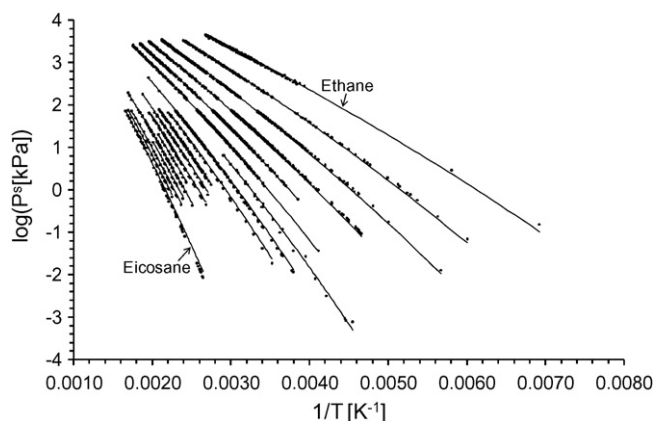


Fig. 1. Series plot of vapor pressure data of *n*-alkanes and their estimation from this work.

bon compound with only one alcohol group), was used to account for the slight “bowing” of the vapor pressure curve at low pressures:

$$\log \left(\frac{p^S}{1 \text{ atm}} \right) = (4.1012 + \text{dB}) \left(\frac{T_{\text{rb}} - 1}{T_{\text{rb}} - 1/8} \right) + \frac{0.75407}{1 + 7 \exp(201/T - 91)} - 1 \quad (9)$$

The results and quality analysis for oxygen, nitrogen and sulphur compounds are presented in Tables 8–10, respectively.

In the case of halogenated compounds, the definition of mono-functional compounds was modified. In this case, the frequency for any particular halogen can be greater than one. The reason for this modification is that for halogenated compounds, no group interaction is considered. Although some halogenated compounds may act as hydrogen-bonding acceptors, the halogen groups can be considered to behave additively due to the weakness of these bonds. The results and quality analysis for halogen and various other elemental compounds are presented in Tables 11 and 12, respectively.

3.4. Multi-functional compounds

The second-order group interactions account for the flawed assumption that all groups are additive. This assumption is not valid for strongly associating groups exhibiting hydrogen-bonding interaction. Increasing the number of these groups leads to competitive association. As with the previous method, a group interaction contribution for non-additive groups is adopted in this study. The list of groups considered to be non-additive is presented in previous works [10,11]. Eq. (7) was derived to calculate the group interaction (GI) contribution. The results and quality analysis for multi-functional compounds are presented in Table 13.

4. Data base preparation

Normal boiling temperatures and experimental vapor pressure data were taken from the Dortmund Data Bank (DDB), the Handbook of Chemistry and Physics and the Beilstein database. The DDB contains approximately 178,000 solid and liquid vapor pressure points from 7160 references, for more than 7400 components (in many cases, however, only one data point may be available for a component). Altogether normal boiling point values for 17,000 components were available. During this study, the available data were carefully checked and numerous questionable data were not considered in the model development. The final regression set

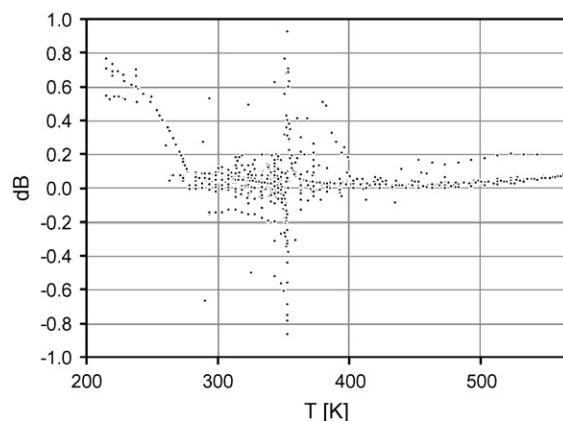


Fig. 2. dB-values for benzene derived from experimental vapor pressure data as a function of temperature.

of data included 68,835 data points for 1663 components which includes 150 points obtained from the Beilstein database.

In a first step, reliable normal boiling points were compiled for all components and then used for the calculation of a dB-value for each data point. When plotted as a function of temperature, the dB-values should scatter around a constant value. Fig. 2 shows a typical set of dB-values obtained for benzene.

Near the normal boiling temperature, dB-values strongly scatter or even diverge systematically. This results from the strong effect of small deviations on the slope parameter that is calculated from two close temperatures. At higher temperature, the vapor pressure equation is not applicable. The steep increase below approximately 280 K indicates solid vapor pressure data below the melting point. dB-values were calculated from vapor pressure points for approx. 1500 components, for which 20 or more data points were available. Unreliable data were removed.

5. Discussion

The results for the estimation of vapor pressures and the Antoine equation for all data points from this work are presented in Table 14. Overall, the proposed method yields results that are in comparable accuracy to the correlative model and applicable to a greater range of compounds. The proposed method shows an excellent agreement between estimated and experimental data and no significant prediction failure was observed.

The distinct advantage for the proposed method as compared to any correlative model is that the estimations are based on functional groups. This implies that the estimation of the slope for any compound was derived from a larger number of chemically similar components. In case of correlative models, the slope depends entirely on the data for the individual component, and thus, on the reliability of the experimental data. As an example, Fig. 3 shows the estimated and predicted vapor pressure curves of arsine compounds. In nearly all cases, there is an excellent agreement between the estimated values and experimental data. The only exception is arsenic sec-butyl dichloride. However, the unusual curvature observed for this compound is not found in case of the other five similar members of this chemical family and consequently, it can be assumed that these data are questionable.

As the proposed method only estimates the slope of the vapor pressure curve, knowledge of a boiling point at a given pressure is required for the calculation of vapor pressure as function of temperature. In this work, a carefully verified normal boiling temperature was used for all estimation, but any other vapor pressure data point can be used to estimate the normal boiling point using Eq. (6).

Table 8
Vapor pressure average relative deviations of this work compared to the Antoine equation for the different types of oxygen compounds (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
Primary alcohols	31	71.0 ¹²²	18.1 ¹²⁷⁵	5.6 ¹⁵⁵⁴	19.0 ¹⁷⁷	14.0 ³¹²⁸	31	37.7 ¹²²	6.0 ¹²⁷⁵	2.2 ¹⁵⁵⁴	17.1 ¹⁷⁷	6.0 ³¹²⁸
Secondary alcohols	30	50.9 ¹³	19.4 ³⁹⁸	3.8 ⁴⁸⁸	6.0 ²⁶	11.2 ⁹²⁵	30	129.2 ¹³	19.2 ³⁹⁸	2.2 ⁴⁸⁸	13.2 ²⁶	11.6 ⁹²⁵
Tertiary alcohols	20	33.5 ²	12.7 ¹⁵³	4.1 ³²¹	13.8 ¹⁸	7.2 ⁴⁹⁴	20	161.6 ²	25.2 ¹⁵³	2.0 ³²¹	5.5 ¹⁸	9.9 ⁴⁹⁴
Aromatic alcohols	20	31.9 ¹³	34.4 ²⁸⁴	3.5 ³⁴⁰	2.2 ⁴⁸	16.7 ⁶⁸⁵	20	44.7 ¹³	9.0 ²⁸⁴	1.7 ³⁴⁰	6.3 ⁴⁸	5.8 ⁶⁸⁵
All alcohols ^a	177	55.2 ²⁶⁰	19.1 ³⁰⁰²	5.5 ³⁸³⁴	16.5 ⁴⁴⁴	13.3 ⁷⁵⁴⁰	171	47.9 ²⁵⁹	10.7 ²⁹⁷¹	3.8 ³⁸²³	18.1 ⁴⁴⁴	8.9 ⁷⁴⁹⁷
Ethers	47	29.5 ¹⁶	6.4 ³¹⁷	2.0 ¹⁶⁴⁸	4.0 ⁴⁴⁶	3.1 ²⁴²⁷	45	63.9 ¹⁶	9.7 ³¹⁰	0.9 ¹⁶³⁷	5.6 ⁴⁴⁶	3.3 ²⁴⁰⁹
Epoxides	8	–	9.3 ²³	2.6 ¹⁷¹	2.1 ³⁷	3.2 ²³¹	7	–	7.1 ²³	1.4 ¹⁶⁹	4.0 ³⁷	2.4 ²²⁹
Aldehydes	22	20.9 ⁴	10.9 ²⁹²	2.6 ⁴⁰³	5.2 ²⁰	6.2 ⁷¹⁹	22	37.6 ⁴	15.2 ²⁹²	1.9 ⁴⁰³	6.2 ²⁰	7.6 ⁷¹⁹
Ketones	50	43.8 ³⁷	8.8 ⁶²⁰	2.2 ¹⁴⁶³	5.9 ²⁶³	5.0 ²³⁸³	48	22.5 ³⁷	5.1 ⁶¹⁵	2.1 ¹⁴⁶¹	7.7 ²⁶³	3.8 ²³⁷⁶
Non-cyclic carbonates	4	33.4 ¹	12.8 ¹⁴⁴	2.1 ²⁴⁹	14.0 ¹⁰	6.3 ⁴⁰⁴	4	28.8 ¹	0.7 ¹⁴⁴	1.0 ²⁴⁹	11.6 ¹⁰	1.2 ⁴⁰⁴
Carboxylic acids	27	85.9 ⁹⁷	18.1 ¹⁸⁰²	6.3 ⁹³⁹	24.0 ¹⁴⁵	16.9 ²⁹⁸³	26	39.2 ⁹⁷	9.1 ¹⁷⁹⁶	3.3 ⁹²⁴	5.0 ¹⁴⁵	8.1 ²⁹⁶²
Esters	78	23.2 ⁵⁰	9.6 ¹²⁰⁹	2.5 ²¹³⁴	6.0 ²⁶⁷	5.4 ³⁶⁶⁰	71	60.6 ⁴⁸	6.8 ¹¹⁴³	2.2 ²⁰⁹³	8.2 ²⁶⁶	5.0 ³⁵⁵⁰
Formic acids esters	12	–	4.6 ⁸¹	1.6 ⁴¹⁸	3.4 ⁵¹	2.2 ⁵⁵⁰	12	–	3.4 ⁸¹	2.5 ⁴¹⁸	7.0 ⁵¹	3.0 ⁵⁵⁰
Lactones	2	–	13.5 ⁴⁷	8.5 ⁴⁴	15.8 ¹	11.2 ⁹²	1	–	19.0 ²²	8.5 ⁴¹	12.0 ¹	12.2 ⁶⁴
Anhydride chains	3	–	9.8 ²⁸	2.9 ³⁴	–	6.0 ⁶²	2	–	6.3 ²⁴	2.9 ³²	–	4.4 ⁵⁶
Anhydride cyclic	2	–	3.4 ³⁵	4.0 ⁷	–	3.5 ⁴²	2	–	5.4 ³⁵	0.7 ⁷	–	4.6 ⁴²
Aromatic oxygen	4	8.7 ¹	7.5 ⁶⁷	4.3 ⁷⁰	4.9 ⁴⁹	5.6 ¹⁸⁷	4	112.1 ¹	39.2 ⁶⁷	4.8 ⁷⁰	4.7 ⁴⁹	17.7 ¹⁸⁷
All (w/o alcohols) ^a	457	42.3 ³⁵³	12.3 ⁶¹³⁰	3.2 ⁹⁶¹²	7.0 ¹⁹²⁰	7.5 ¹⁸⁰¹⁵	432	45.7 ³⁴⁷	8.5 ⁵⁹⁷⁶	2.7 ⁹⁴⁹⁵	8.1 ¹⁹¹⁹	6.1 ¹⁷⁷³⁷

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— 0.01 kPa $\leq P \leq 10$ kPa, MP— 10 kPa $< P \leq 500$ kPa, HP— $P > 500$ kPa.

^a Includes multi-functional compounds.

Table 9
Vapor pressure average relative deviations of this work compared to the Antoine equation for the different types of nitrogen compounds (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
Primary amines	36	24.6 ²	16.3 ⁴⁸⁵	3.3 ⁶⁵²	10.5 ¹⁰³	9.0 ¹²⁴²	35	186.8 ¹	8.0 ⁴⁷⁹	1.9 ⁶⁵¹	11.3 ¹⁰³	5.2 ¹²³⁴
Secondary amines	22	–	10.9 ¹⁶⁵	2.8 ⁴⁶³	5.3 ¹¹⁵	5.0 ⁷⁴³	22	–	4.4 ¹⁶⁵	1.3 ⁴⁶³	4.9 ¹¹⁵	2.6 ⁷⁴³
Tertiary amines	12	–	14.4 ²¹⁵	4.5 ²⁷⁴	3.5 ²⁹	8.6 ⁵¹⁸	12	–	11.4 ²¹⁵	2.0 ²⁷⁴	6.6 ²⁹	6.1 ⁵¹⁸
All amines	109	30.2 ²⁹	14.5 ¹²³⁵	4.0 ¹⁸⁹¹	9.7 ³⁰⁰	8.4 ³⁴⁵⁵	104	55.0 ²⁵	10.7 ¹²²⁴	2.2 ¹⁸⁶⁸	8.3 ³⁰⁰	6.1 ³⁴¹⁷
Five-membered N rings	8	–	18.6 ¹⁰⁴	6.1 ¹⁷⁹	7.3 ¹⁵	10.5 ²⁹⁸	7	–	2.3 ⁸⁶	0.7 ¹⁶²	2.0 ¹⁵	1.3 ²⁶³
Six-membered N rings	25	–	4.8 ⁴²⁵	1.6 ⁸⁸⁸	3.2 ⁶⁷	2.7 ¹³⁸⁰	25	–	4.4 ⁴²⁵	1.1 ⁸⁸⁸	4.2 ⁶⁷	2.3 ¹³⁸⁰
Cyanides	25	15.8 ¹⁵	6.8 ⁴⁰⁵	2.5 ³⁷⁸	5.2 ³¹	4.9 ⁸²⁹	25	85.0 ¹⁵	8.9 ⁴⁰⁵	1.8 ³⁷⁸	14.7 ³¹	7.3 ⁸²⁹
Amides	1	–	3.0 ⁵	0.0 ¹	–	2.5 ⁶	1	–	5.6 ⁵	0.0 ¹	–	4.7 ⁶
Mono amides	4	–	9.1 ¹⁸⁵	5.8 ⁹¹	1.1 ²	8.0 ²⁷⁸	4	–	5.9 ¹⁸⁵	3.0 ⁹¹	20.0 ²	5.0 ²⁷⁸
Di amides	7	14.6 ¹	12.4 ³¹⁶	3.5 ²⁸³	–	8.2 ⁶⁰⁰	7	0.0 ¹	8.1 ³¹⁶	3.4 ²⁸³	–	5.9 ⁶⁰⁰
Iso cyanates	9	–	19.1 ⁴⁶	5.3 ⁷⁵	–	10.5 ¹²¹	9	–	9.0 ⁴⁶	4.5 ⁷⁵	–	6.2 ¹²¹
Oximes	6	13.4 ²	4.0 ⁵⁶	1.9 ⁴⁰	–	3.3 ⁹⁸	6	12.1 ²	3.8 ⁵⁶	1.0 ⁴⁰	–	2.8 ⁹⁸
Nitrous and nitrite	9	9.9 ⁴	7.3 ¹⁸⁵	3.0 ¹³⁶	–	5.5 ³²⁵	8	31.2 ⁴	6.6 ¹⁸⁵	2.3 ¹³⁴	–	5.1 ³²³
Nitrates	3	–	16.0 ¹⁶	6.5 ¹⁴	–	11.6 ³⁰	3	–	1.1 ¹⁶	0.3 ¹⁴	–	0.7 ³⁰
N–C ₄	3	–	11.1 ¹⁰	6.6 ¹⁶	–	8.3 ²⁶	3	–	5.9 ¹⁰	2.3 ¹⁶	–	3.7 ²⁶
All (w/o amines) ^a	120	28.7 ²³	9.5 ¹⁸⁷⁴	3.0 ²¹⁷⁴	4.3 ¹¹⁵	6.1 ⁴¹⁸⁶	116	66.8 ²³	6.7 ¹⁸³⁹	1.9 ²¹³⁷	7.0 ¹¹⁵	4.5 ⁴¹¹⁴

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— 0.01 kPa $\leq P \leq 10$ kPa, MP— 10 kPa $< P \leq 500$ kPa, HP— $P > 500$ kPa.

^a Includes multi-functional compounds.

For the training set employed in this work, the back calculation of the normal boiling point from vapor pressure data produced an average absolute deviation of 8.9 K for extremely low vapor pressures (ELPs), 2.8 K for low vapor pressures (LPs), 1.2 K for moderate

pressures (MPs), 2.7 K for HPs and 2.1 K for all data points. With the exception of the extremely low vapor pressure data set, the reported deviations for the back calculation of the normal boiling temperature are well within the deviations in temperature pre-

Table 10
Vapor pressure average relative deviations of this work compared to the Antoine equation for the different types of sulphur compounds (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
Disulfides	4	–	2.5 ⁷⁴	1.7 ¹³⁰	–	2.0 ²⁰⁴	4	–	0.9 ⁷⁴	0.2 ¹³⁰	–	0.5 ²⁰⁴
Thiols	30	89.0 ⁴	4.5 ⁸⁴	2.3 ⁵⁷⁴	10.3 ⁶⁴	3.8 ⁷²⁶	29	0.6 ¹	1.2 ⁸⁴	0.4 ⁵⁷³	2.8 ⁶⁴	0.7 ⁷²²
Thioether	14	–	9.2 ⁵⁸	1.3 ³⁰⁴	5.2 ²²	2.7 ³⁸⁴	14	–	5.6 ⁵⁸	0.7 ³⁰⁴	5.9 ²²	1.7 ³⁸⁴
Aromatic thioether	7	–	9.6 ¹⁰⁴	3.1 ¹⁹⁹	6.9 ³⁵	5.5 ³³⁸	7	–	4.3 ¹⁰⁴	1.2 ¹⁹⁹	3.9 ³⁵	2.5 ³³⁸
Sulfolane (O=S=O)	2	32.6 ⁵	9.4 ²²	0.6 ¹³	–	9.4 ⁴⁰	1	30.4 ⁵	7.0 ²⁰	1.7 ¹²	–	8.5 ³⁷
Isothiocyanates	3	–	4.0 ⁴⁸	2.6 ⁵	–	3.9 ⁵³	3	–	1.2 ⁴⁸	10.4 ⁵	–	2.0 ⁵³
Sulfates, sulfon amides and sulfoxides	5	–	9.1 ³⁸	0.0 ⁵	–	8.0 ⁴³	2	–	11.9 ¹³	3.4 ²	–	10.8 ¹⁵
All compounds ^a	99	40.4 ²¹	9.0 ⁵⁸¹	2.1 ¹⁶²⁴	8.4 ¹²¹	4.5 ²³⁴⁷	94	16.7 ¹⁸	4.0 ⁵⁵⁴	0.7 ¹⁶¹⁹	3.7 ¹²¹	1.8 ²³¹²

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— 0.01 kPa $\leq P \leq 10$ kPa, MP— 10 kPa $< P \leq 500$ kPa, HP— $P > 500$ kPa.

^a Includes multi-functional compounds.

Table 11

Vapor pressure average relative deviations of this work compared to the Antoine equation for the different types of halogenated compounds (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
Fluorine saturated	54	–	8.7 ²⁴⁵	2.3 ¹⁵⁷⁸	6.3 ²⁷³⁸	5.1 ⁴⁵⁶¹	53	–	6.5 ²⁴⁴	1.5 ¹⁵⁷⁷	2.0 ²⁷³⁸	2.1 ⁴⁵⁵⁹
All fluorine	75	–	7.2 ³⁶¹	2.4 ²¹⁵³	6.0 ²⁹⁹³	4.7 ⁵⁵⁰⁷	74	–	4.9 ³⁶⁰	1.3 ²¹⁵²	2.4 ²⁹⁹³	2.1 ⁵⁵⁰⁵
Chlorine saturated	49	16.4 ⁴	6.9 ³⁶¹	2.5 ¹⁵⁷⁴	6.9 ⁴⁶⁸	4.0 ²⁴⁰⁷	49	60.2 ⁴	6.3 ³⁶¹	1.5 ¹⁵⁷⁴	4.7 ⁴⁶⁸	2.9 ²⁴⁰⁷
All chlorine	81	10.5 ¹¹	6.8 ⁷³³	2.5 ²⁴⁷¹	6.0 ⁵⁷⁹	3.9 ³⁷⁹⁴	80	33.3 ¹¹	5.3 ⁷³²	1.5 ²⁴⁷⁰	4.5 ⁵⁷⁹	2.8 ³⁷⁹²
Bromine saturated	18	21.0 ¹⁰	8.5 ¹⁵⁰	2.6 ¹⁹⁴	–	5.6 ³⁵⁴	18	5.9 ¹⁰	5.1 ¹⁵⁰	2.2 ¹⁹⁴	–	3.5 ³⁵⁴
All bromine	31	32.5 ¹⁶	9.1 ³⁵⁷	2.4 ⁴⁴¹	2.2 ¹⁵	5.9 ⁸²⁹	30	35.1 ¹⁶	6.4 ³⁵⁵	3.2 ⁴³⁰	2.9 ¹⁵	5.2 ⁸¹⁶
All iodine	10	–	7.6 ¹¹⁰	2.1 ¹⁵³	3.8 ⁵	4.4 ²⁶⁸	10	–	4.6 ¹¹⁰	1.4 ¹⁵³	2.3 ⁵	2.7 ²⁶⁸
All compounds ^a	353	28.0 ³⁴	9.1 ²⁴⁰¹	2.7 ⁷⁴⁵⁶	5.5 ⁵⁵⁷⁰	4.7 ¹⁵⁴⁶¹	343	33.7 ³⁴	7.7 ²³⁵⁸	1.8 ⁷⁴²⁵	3.7 ⁵⁵⁶⁹	3.5 ¹⁵³⁸⁶

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— $0.01 \text{ kPa} \leq P \leq 10$ kPa, MP— $10 \text{ kPa} < P \leq 500$ kPa, HP— $P > 500$ kPa.

^a Includes multi-functional compounds.

Table 12

Vapor pressure average relative deviations of this work compared to the Antoine equation for the different types of other elemental compounds (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
Phosphates	3	15.8 ⁷	18.6 ⁸	5.6 ⁸	–	13.2 ²³	3	9.6 ⁷	6.1 ⁸	2.0 ⁸	–	5.7 ²³
Arsine	6	–	10.8 ¹⁵	2.1 ²²	–	5.6 ³⁷	6	–	22.3 ¹⁵	1.5 ²²	–	9.9 ³⁷
Germanium	1	–	8.5 ²²	1.7 ¹³	–	5.9 ³⁵	1	–	4.8 ²²	3.0 ¹³	–	4.1 ³⁵
Germanium—(Cl) ₃	3	–	1.1 ³³	0.5 ²⁶	–	0.9 ⁵⁹	1	–	1.2 ¹⁵	0.2 ⁸	–	0.9 ²³
Tin	3	–	1.7 ⁵⁵	0.5 ⁹⁰	2.0 ¹⁴	1.0 ¹⁵⁹	3	–	1.5 ⁵⁵	0.6 ⁹⁰	7.2 ¹⁴	1.5 ¹⁵⁹
Boron	6	–	8.0 ²⁶	3.3 ⁶⁹	–	4.6 ⁹⁵	6	–	6.8 ²⁶	2.5 ⁶⁹	–	3.7 ⁹⁵
Silicon	23	–	6.7 ¹²³	2.4 ²⁵⁸	1.7 ²⁸	3.7 ⁴⁰⁹	21	–	1.7 ¹¹⁵	0.6 ²⁵²	1.7 ²⁸	1.0 ³⁹⁵
Silicon connected to O,F or Cl	65	–	11.2 ²⁷²	2.7 ⁶³¹	2.7 ¹⁴⁰	4.9 ¹⁰⁴³	65	–	6.3 ²⁷²	1.7 ⁶³¹	11.9 ¹⁴⁰	4.3 ¹⁰⁴³
Acid chloride	2	–	15.6 ⁶⁷	6.9 ⁹	–	14.6 ⁷⁶	2	–	3.6 ⁶⁷	30.5 ⁹	–	6.8 ⁷⁶
Urea	1	–	1.4 ¹⁴	0.7 ¹³	–	1.0 ²⁷	1	–	0.6 ¹⁴	0.9 ¹³	–	0.8 ²⁷
Selenium	1	–	–	1.6 ²⁸	–	1.6 ²⁸	1	–	–	1.8 ²⁸	–	1.8 ²⁸
Aluminium	1	–	3.8 ²⁰	1.0 ¹⁴	–	2.7 ³⁴	1	–	0.3 ²⁰	0.1 ¹⁴	–	0.2 ³⁴

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— $0.01 \text{ kPa} \leq P \leq 10$ kPa, MP— $10 \text{ kPa} < P \leq 500$ kPa, HP— $P > 500$ kPa. *Includes multi-functional compounds.

Table 13

Vapor pressure average relative deviations of this work compared to the Antoine equation for the different types of multi-functional compounds (number of points in superscript)

	This work—average relative deviation (%)						Antoine—average relative deviation (%)					
	NC	ELP	LP	MP	HP	AV	NC	ELP	LP	MP	HP	AV
OH	54	47.1 ⁹⁴	17.3 ⁷⁵²	6.7 ⁸¹⁷	19.1 ⁵⁷	13.9 ¹⁷²⁰	51	48.3 ⁹³	11.4 ⁷⁴⁵	8.8 ⁸¹⁴	51.9 ⁵⁷	13.5 ¹⁷⁰⁹
OH(a)	6	7.3 ¹⁶	5.7 ³⁵	1.9 ²⁴	–	4.8 ⁷⁵	6	45.4 ¹⁶	26.2 ³⁵	28.3 ²⁴	–	31.0 ⁷⁵
NH ₂	14	4.9 ³	11.0 ¹⁴⁴	6.4 ²⁵⁹	20.0 ⁵³	9.4 ⁴⁵⁹	12	–	8.8 ¹³⁹	4.2 ²⁵⁶	9.6 ⁵³	6.3 ⁴⁴⁸
NH	8	51.8 ¹⁴	16.3 ¹⁰⁹	2.6 ¹⁵⁴	20.0 ²⁷	11.3 ³⁰⁴	5	51.4 ¹⁴	11.3 ¹⁰⁸	1.6 ¹³³	9.1 ²⁷	8.5 ²⁸²
SH	10	–	9.5 ¹³	2.1 ²⁸	–	4.5 ⁴¹	10	–	0.2 ¹³	0.6 ²⁸	–	0.5 ⁴¹
COOH	3	–	3.8 ⁷	0.3 ⁵	–	2.4 ¹²	3	–	7.4 ⁷	2.0 ⁵	–	5.1 ¹²
OCN	3	92.9 ²	17.5 ³⁶	4.8 ²⁷	–	14.6 ⁶⁵	3	38.6 ²	5.0 ³⁶	6.6 ²⁷	–	6.7 ⁶⁵
Ether	91	13.0 ³¹	9.1 ⁶⁹⁷	3.9 ¹¹⁶⁹	8.1 ¹⁸²	6.2 ²⁰⁷⁹	85	52.4 ³⁰	8.9 ⁶⁷⁵	5.1 ¹¹⁴¹	10.1 ¹⁸²	7.5 ²⁰²⁸
Epoxide	3	–	3.7 ³¹	1.7 ¹⁵	–	3.0 ⁴⁶	3	–	3.9 ³¹	2.7 ¹⁵	–	3.5 ⁴⁶
Ester	40	24.2 ¹¹⁶	16.0 ³⁷⁰	7.5 ³⁸³	14.6 ⁷⁹	13.5 ⁹⁴⁸	36	48.0 ¹¹³	9.9 ³⁵⁷	6.1 ³⁶⁹	11.6 ⁷⁹	13.2 ⁹¹⁸
Ketone	11	2.1 ²	10.0 ⁶⁴	6.9 ¹²⁹	16.1 ⁴²	9.3 ²³⁷	11	37.3 ²	9.8 ⁶⁴	5.9 ¹²⁹	15.7 ⁴²	8.9 ²³⁷
Aldehyde	3	–	7.5 ⁵³	1.7 ⁵⁰	0.7 ²	4.6 ¹⁰⁵	3	–	6.0 ⁵³	2.6 ⁵⁰	11.9 ²	4.5 ¹⁰⁵
Nitro, AtS, CN, AO, AN5, AN6	20	89.8 ²	11.5 ²¹⁹	3.7 ¹⁷⁷	0.7 ²	8.4 ⁴⁰⁰	17	5.2 ²	8.4 ²⁰²	3.1 ¹⁵⁷	11.9 ²	6.1 ³⁶³
All GI components	199	32.9 ²³²	13.9 ¹⁸⁴²	5.2 ²³⁸⁵	13.0 ³⁵³	10.4 ⁴⁸¹²	185	47.5 ²²⁸	9.8 ¹⁷⁹⁹	5.4 ²³²⁰	17.4 ³⁵³	10.1 ⁴⁷⁰⁰

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— $0.01 \text{ kPa} \leq P \leq 10$ kPa, MP— $10 \text{ kPa} < P \leq 500$ kPa, HP— $P > 500$ kPa. *Includes multi-functional compounds.

Table 14

Vapor pressure average relative deviations for all compounds

	Average absolute, relative deviation (number of points in superscript)					
	NC	ELP	LP	MP	HP	AV
This work (K)	1663	3.1 ¹⁰³⁰	1.8 ²⁰²⁴³	0.8 ³⁵⁹⁵²	3.2 ¹¹⁶¹⁰	1.6 ⁶⁸⁸³⁵
This work (% kPa)	1663	40.4 ¹⁰³⁰	11.0 ²⁰²⁴³	2.8 ³⁷⁶¹⁵	5.7 ¹¹⁶¹⁰	6.2 ⁷⁰⁴⁹⁸
Antoine (% kPa)	1603	43.7 ¹⁰¹⁷	7.6 ¹⁹⁹²⁹	1.8 ³⁷³⁷⁴	4.8 ¹¹⁶⁰⁷	4.5 ⁶⁹⁹²⁷

NC, number of components; AV, average relative deviations for all points. The following abbreviations denotes pressure ranges, ELP— $P < 0.01$ kPa, LP— $0.01 \text{ kPa} \leq P \leq 10$ kPa, MP— $10 \text{ kPa} < P \leq 500$ kPa, HP— $P > 500$ kPa. *Includes multi-functional compounds.

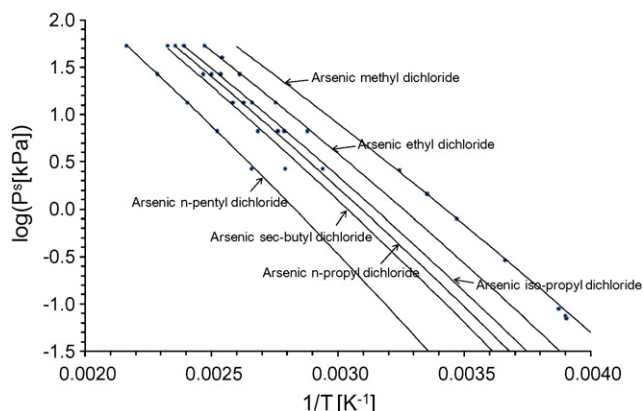


Fig. 3. Multiple plot of vapor pressure data ($\log(P)$ vs. $1/T$) of arsine compounds and their estimation from this work.

sented in Table 14. The slightly higher deviations for the estimation of extremely low vapor pressures, as well as, the back calculation of the normal boiling point for extremely low vapor pressures (<0.01 kPa) is to a great extent due to the poorer quality of these data points.

To test the predictive capability of the vapor pressure estimation method, two test sets of data not used in the regression were prepared.

The first test set consisted of 1978 data points for 396 components for which only a few low-pressure data points were available. Because of the insufficient means to verify these data, they were usually not included into the training set. For this set of components, the proposed method yielded an average absolute deviation of 7% in vapor pressure compared to 6.2% for 70,498 data points used for parameter regression (Table 14).

In order to verify the correction term for mono-functional alcohols (Eq. (9)), six of these components (1982 data points) were removed from the regression set. For this set of data, the proposed method yielded an average absolute deviation of 12.1% in vapor pressure compared to 13.3% for 7540 data points in the training set (Table 8).

In some cases, only data for a single component were available to regress a group contribution. Although a single reliable data point is in principal a sufficient basis for a group parameter, data from different sources and for different components increase the reliability of the parameter. To avoid unrealistic predictions, group contribution values were compared to other groups of similar chemical nature and for which a larger set of data was available. Most values were found to be acceptable, however, two groups, ID–91¹ and ID–187,² revealed improbable contribution values and should be used with caution.

6. Conclusion

A new group contribution method has been developed for the prediction of liquid vapor pressure as a function of temperature. The method requires knowledge of only the molecular structure and the normal boiling point (or any other vapor pressure point at not too high pressure). Overall, the proposed method yielded results that are in comparable accuracy to the Antoine equation with previously regressed parameters and shows an excellent agreement between estimated and experimental data with no apparent probability of prediction failure. The method finds most useful application in the

low-pressure region. Only components with reliable normal boiling temperature information were used for parameter regression. Model parameters generally show physically realistic values.

List of symbols

a, b, c	adjustable parameters
c_{ij}	group interaction contribution between group i and group j [K] ($c_{ii} = 0$)
C	group contribution
dB	slope parameter
GI	total group interaction contribution
H	enthalpy
$\log$	logarithm (base 10)
n	number of atoms except hydrogen
N	number of groups
P	pressure
S	entropy
T	absolute temperature [K]
V	volume
z	compressibility factor

Subscripts

b	boiling
c	critical
vap	vaporization

Superscripts

s	saturation
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Appendix A. Calculation of further properties from the original model equation

Boiling temperature	$T = \frac{(1/8)\log(P^s/1 \text{ atm}) - (4.1012 + dB)}{\log(P^s/1 \text{ atm}) - (4.1012 + dB)} T_b$
Calculation of dB from ΔH_{vap}	$dB = \frac{-\Delta_{vap}H((T_b/T) - 8)^2}{56 R T_b \Delta_{vap} z \ln(10)} - 4.1012$
Enthalpy of vaporization	$\Delta_{vap}H = \frac{-56(4.1012 + dB)RT_b \Delta_{vap} z \ln(10)}{((T_b/T) - 8)^2}$

Appendix B. Examples

To illustrate the application of the proposed method, a detailed procedure for the estimation of vapor pressure is given in Tables 15a–15h for

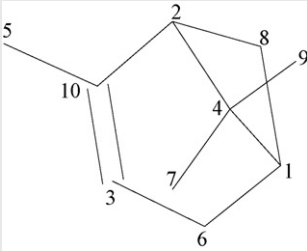
- Alpha-pinene illustrating the application of the steric correction.
- 1,2-Ethanediol illustrating the application of group interactions.
- Perfluoro-2-propanone illustrating the application of halogen corrections.
- Acrylic acid illustrating the application of the $C=C-C=O$ correction.
- Glycol monoacetate illustrating the application of group interactions for two different interacting groups.
- Dipropyl succinate illustrating the application of group interactions and the calculation of the normal boiling point from an experimental vapor pressure data point.
- Diethanolamine illustrating the application of group interactions in case of three interacting groups.

¹ –O– connected to two neighbors which are each either C or Si (ethers).

² Epoxide–epoxide interaction parameter.

Table 15a

Estimation of vapor–liquid equilibrium curve slope (dB) of alpha-pinene and liquid vapor pressure at 388.15 K

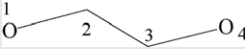
Component: alpha-pinene, number of atoms: 10, formula: C ₁₀ H ₁₆				
Group	Atoms	Frequency	Contribution	Total
1	5, 7, 9	3	0.0133063	0.0399189
9	6, 8	2	0.0222573	0.0445146
10	1, 2	2	0.0328162	0.0656324
11	4	1	0.0048500	0.0048500
62	3–10	1	0.0974210	0.0974210
Corrections				
125	1–4–2–8	1	0.0339765	0.0339765
132	2–4, 1–4	2	–0.0011994	–0.0023988
Total sum				0.2839146

$$dB = 0.2839146 - 0.176055 = 0.1078596$$

$$T_{b,exp} = 429.00 \text{ K}$$

$$p^S = \left[10^{(4.1012 + 0.1078596) \left(\frac{388.15 \text{ K}}{429.00 \text{ K}} - 1 \right) / \left(\frac{388.15 \text{ K}}{429.00 \text{ K}} - (1/8) \right)} \right] \times 101.325 \text{ kPa} = 31.03 \text{ kPa} \quad p_{exp}^S = 31.68 \text{ kPa}$$

Table 15bEstimation of vapor–liquid equilibrium curve slope (dB) of 1,2-ethanediol and liquid vapor pressure at 410.65 K^a

Component: 1,2-ethanediol, number of atoms: 4, formula: HOCH ₂ CH ₂ OH				
Group	Atoms	Frequency	Contribution	Total
7	2, 3	2	0.0378485	0.0756970
36	1, 4	2	0.7560824	1.5121648
Corrections				
135	1–4	2	–0.5615153	–0.2807577
Total sum				1.3071041

$$GI = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 \frac{C_{OH-OH}}{2-1} = \frac{1}{2} C_{OH-OH} = -0.2807577$$

Number of atoms except hydrogen: 2,

$$dB = 1.3071041 - 0.176055 = 1.1310491$$

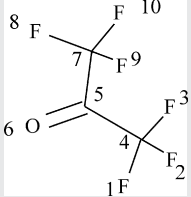
$$T_{b,exp} = 470.50 \text{ K}$$

$$p^S = \left[10^{(4.1012 + 1.1310491) \left(\frac{410.65 \text{ K}}{470.50 \text{ K}} - 1 \right) / \left(\frac{410.65 \text{ K}}{470.50 \text{ K}} - (1/8) \right)} \right] \times 101.325 \text{ kPa} = 13.05 \text{ kPa} \quad p_{exp}^S = 12.27 \text{ kPa}$$

^a The double sum adds up the interaction of the first with the second and the second with the first OH-group (which are both equal to –0.5615153) as well as the interactions of the first OH-group with itself and the second OH-group with itself (which are both zero).

Table 15c

Estimation of vapor–liquid equilibrium curve slope (dB) of perfluoro-2-propanone and liquid vapor pressure at 210.16 K

Component: perfluoro-2-propanone, number of atoms: 10, formula: (CO)(CF ₃) ₂				
Group	Atoms	Frequency	Contribution	Total
7	4, 7	2	0.0378485	0.0756970
21	1, 2, 3, 8, 9, 10	6	0.0420887	0.2525322
51	5, 6	1	0.255848	0.2558480
Corrections				
120	5	1	0.002503	0.0025030
121	4, 7	2	–0.0833326	–0.1666652
123	–	1	–0.1259208	–0.1259208
Total Sum				0.2939942

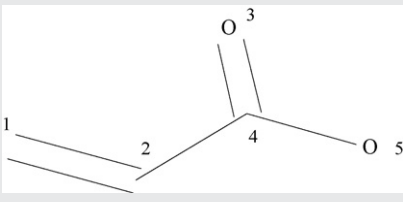
$$dB = 0.2939942 - 0.176055 = 0.1179392$$

$$T_{b,exp} = 245.90 \text{ K}$$

$$p^S = \left[10^{(4.1012 + 0.1179392) \left(\frac{210.16 \text{ K}}{245.90 \text{ K}} - 1 \right) / \left(\frac{210.16 \text{ K}}{245.90 \text{ K}} - (1/8) \right)} \right] \times 101.325 \text{ kPa} = 14.63 \text{ kPa} \quad p_{exp}^S = 14.05 \text{ kPa}$$

Table 15d

Estimation of vapor–liquid equilibrium curve slope (dB) of acrylic acid and liquid vapor pressure at 344.15 K

Component: acrylic acid, number of atoms: 5, formula: $\text{CH}_2=\text{CHCOOH}$				
Group	Atoms	Frequency	Contribution	Total
44	4-(3, 5)	1	1.0741000	1.0741000
61	1-2	1	0.0342541	0.0342541
Corrections				
134	1-2-4-3	1	-0.0159694	-0.0159694
Total sum				1.0923847

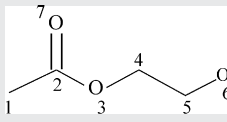
$$\text{dB} = 0.2939942 - 1.0923847 = 0.9163297$$

$$T_{\text{b,exp}} = 413.60 \text{ K}$$

$$p^{\text{S}} = \left[10^{(4.1012 + 0.9163297)((344.15 \text{ K}/413.60 \text{ K}) - 1/(344.15 \text{ K}/413.60 \text{ K}) - (1/8))} \right] \times 101.325 \text{ kPa} = 6.52 \text{ kPa} \quad p_{\text{exp}}^{\text{S}} = 6.67 \text{ kPa}$$

Table 15e

Estimation of vapor–liquid equilibrium curve slope (dB) of glycol monoacetate and liquid vapor pressure at 352.65 K

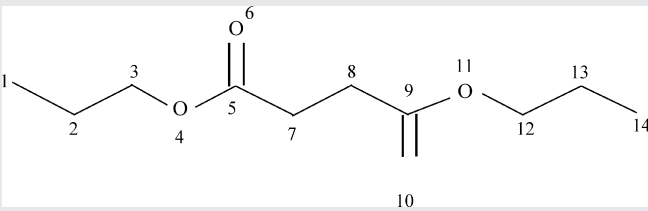
Component: glycol monoacetate, number of atoms: 7, formula: $\text{CH}_3(\text{CO})\text{OCH}_2\text{CH}_2\text{OH}$				
Group	Atoms	Frequency	Contribution	Total
1	1	1	0.0133063	0.0133063
7	4, 5	2	0.0378485	0.0756970
36	6	1	0.7560824	0.7560824
45	2-(3, 7)	1	0.3557381	0.3557381
Total				1.2008238
Corrections				
142	6-(2-(3, 7))	2	-1.7976930	-3.595386

$\text{dB} = 1.2008238 - \frac{1}{7} 3.5953860 - 0.176055 = 0.5111422$. Böhme and Opfer [20] reported a normal boiling temperature of $T_{\text{b,exp}} = 458.65 \text{ K}$ and a vapor pressure of 1.6 kPa at 352.65 K. Using the reported normal boiling point, estimation of the vapor pressure at 352.65 K leads to:

$$p^{\text{S}} = \left[10^{(4.1012 + 0.5111422)((352.65 \text{ K}/458.65 \text{ K}) - 1/(352.65 \text{ K}/458.65 \text{ K}) - (1/8))} \right] \times 101.325 \text{ kPa} = 2.24 \text{ kPa} \quad p_{\text{exp}}^{\text{S}} = 1.6 \text{ kPa}$$

Table 15f

Estimation of vapor–liquid equilibrium curve slope (dB) of dipropyl succinate and the normal boiling temperature from a low vapor pressure data point

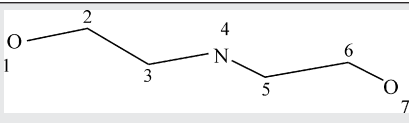
Component: dipropyl succinate, number of atoms: 14, formula: $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}(\text{CO})\text{CH}_2\text{CH}_2(\text{CO})\text{OCH}_2\text{CH}_2\text{CH}_3$				
Group	Atoms	Frequency	Contribution	Total
1	1, 14	2	0.0133063	0.0266126
4	2, 7, 8, 13	4	0.0546564	0.2186256
7	3, 12	2	0.0378485	0.0756970
45	5-(6, 4), 9-(10, 11)	2	0.3557381	0.7114762
Total				1.0324110
Corrections				
189	5-(6, 4)-9-(10, 11)	2	0.9203138	1.8406276

$\text{dB} = 1.0324110 + \frac{1}{14} 1.8406276 - 0.176055 = 0.9878298$. For this component, Vogel [21] reported a vapor pressure of 0.4 kPa at 374.65 K.

$$T_{\text{b}} = \frac{\log(1 \text{ atm}/0.4 \text{ kPa}) - (4.1012 + 0.9878298)}{(1/8) \log(1 \text{ atm}/0.4 \text{ kPa}) - (4.1012 + 0.9878298)} \times 374.65 \text{ K} \quad T_{\text{b,calc}} = 520.85 \text{ K}, \text{ Mustafaev and Ragimov [22] reported a normal boiling temperature of } 523.95 \text{ K}.$$

Table 15g

Estimation of vapor–liquid equilibrium curve slope (dB) of diethanolamine and liquid vapor pressure at 401.13 K

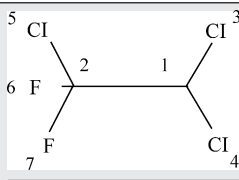
Component: diethanolamine, number of atoms: 7, formula: $\text{NH}(\text{C}_2\text{H}_4\text{OH})_2$				
Group	Atoms	Frequency	Contribution	Total
7	2, 3, 5, 6	4	0.0378485	0.1513940
36	1, 7	2	0.7560824	1.5121648
42	4	1	0.1937667	0.1937667
Total				1.8573255
Corrections				
135	1–7	2	−0.5615153	−1.1230306
137	1–4, 7–4	4	0.0424825	0.1699300
				−0.9531006

$$\text{dB} = 1.8573255 - \frac{1}{7 \times (3-1)} 0.9531006 - 0.176055 = 1.7493490 \quad T_{b,\text{exp}} = 541.15 \text{ K}$$

$$p^S = \left[10^{(4.1012 + 1.7493490)(401.13 \text{ K}/541.15 \text{ K}) - 1/(401.13 \text{ K}/541.15 \text{ K}) - (1/8)} \right] \times 101.325 \text{ kPa} = 0.35421 \text{ kPa} \quad p_{\text{exp}}^S = 0.410 \text{ kPa}$$

Table 15h

Estimation of vapor–liquid equilibrium curve slope (dB) of 1,2,2-trichloro-1,1-difluoroethane [R122] and liquid vapor pressure at 297.46 K

Component: 1,2,2-trichloro-1,1-difluoroethane [R122], number of atoms: 5, formula: $\text{CHCl}_2\text{CClF}_2$				
Group	Atoms	Frequency	Contribution	Total
7	1, 2	2	0.0378485	0.075697
21	6, 7	2	0.0420887	0.0841774
26	3, 4	2	0.0678082	0.1356164
27	5	1	0.0559304	0.0559304
Corrections				
121	2	1	−0.0833326	−0.0833326
124		1	−0.0472962	−0.0472962
Total sum				0.2207924

$$\text{dB} = 0.2207924 - 0.176055 = 0.0447374 \quad T_{b,\text{exp}} = 344.25 \text{ K}$$

$$p^S = \left[10^{(4.1012 + 0.0447374)(297.46 \text{ K}/344.25 \text{ K}) - 1/(297.46 \text{ K}/344.25 \text{ K}) - (1/8)} \right] \times 101.325 \text{ kPa} = 17.5093 \text{ kPa} \quad p_{\text{exp}}^S = 16.804 \text{ kPa}$$

- 1,2,2-Trichloro-1,1-difluoroethane [R122] illustrating the application of group priority.

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